Controlling Shareholders and Supermajority Rules

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November 15, 2025

Motivation I: Controlling Shareholders in Practice

- **Global dominance of controlling shareholders (CS).**
  - Outside the U.S. & UK, concentrated ownership is the dominant shareholding structure.
  - Even in the U.S., dual-class structures and controlling shareholders are increasingly prominent in "superstar" firms (e.g. Tesla, Meta, Alphabet).
- **Recent legal shifts.**
  - Delaware SB 21 represents a major shift in power toward controlling shareholders.
- **Exit without voice.**
  - Absent corporate-law constraints, a CS effectively determines both board and shareholder elections.
  - Minority shareholders often face a choice between *exit* and (essentially) *no voice*; in private companies even exit can be severely constrained.
- **Conflicted transactions.**
  - Related-party transactions or "tunnelling" are a first-order concern where CS are present.

Motivation II: How Law Regulates Conflicted Deals

- **Basic idea:** let minority shareholders vote on conflicted transactions, excluding the CS.
- Two broad voting designs appear across jurisdictions:
  1. **Majority of the Minority (MoM):** more than 50% of minority shares must vote "Yes" for the proposal to pass.
  2. **Supermajority rules:** a required fraction $\pi \in (0, 1]$ of minority shares must vote "Yes" for passage.
     - $\pi = 0.5$ is MoM; $\pi \to 1$ approximates unanimity.
     - If 100% of minority shareholders must agree, the effective rule is unanimity.

Motivation III: Illustrative UK Rules

- **UK Companies Act 2006, s. 283(1):**

  "A special resolution of the members (or of a class of members) of a company means a resolution passed by a majority of not less than 75%."
- **UK Companies Act 2006, s. 21(1):**

  "A company may amend its articles by special resolution."
- **Smith v Croft (No 2) [1988] Ch 114:**

  An applicant for leave to continue a derivative action must show support from a majority of the independent minority.

Motivation IV: Beyond Type I / Type II Errors

- Standard intuition: as the supermajority requirement tightens, **Type I errors** (approving bad projects) fall and **Type II errors** (blocking good projects) rise.
- Loosening the threshold reverses this trade-off.
- **But this is not the only effect.**
  - Changing the voting regime also changes the CS's *incentive to misrepresent* project quality.
- Why?
  - The CS enjoys private benefits when projects are approved, so it has an inherent gain from exaggerating quality.
  - Lying is costly (e.g. litigation risk, enforcement, reputational sanctions), so the CS trades off *benefits of approval* against *signalling costs*.
  - Under unanimity (essentially $\pi = 1$), the CS must persuade *everyone in the room*; when $\pi = 0$, persuasion is not required at all.

Research Questions

- **Idea of the paper:** model *costly information transmission* between a CS and minority shareholders under different voting regimes.
- Key ingredients:
  - CS privately observes the quality of a conflicted project.
  - CS chooses a costly signal (e.g. shading or exaggerating a management proposal).
  - Minorities update to posterior beliefs and vote on the proposal.
- **Main questions:**
  - How large are *pooling regions* under MoM vs. different supermajority thresholds?
  - When do tighter rules *increase* or *decrease* pooling relative to more lenient rules?
- **Welfare implications:**
  - Lying is costly to the CS (truth-telling is free).
  - Misrepresentation can also:
    1. allow bad projects that should be blocked to be approved; and
    2. generate ex post litigation and enforcement costs when misstatements are uncovered.

Headline Results I: Talk-Up Regime

- **Equilibrium structure (talk-up).**
  - Threshold strategies with a belief cut-off (the "bar" u).
  - Partial pooling *below* the bar: controllers shade valuations up to clear the bar, and a range of projects are accepted despite misrepresentation.
  - Types below a marginal threshold $\theta^*(u)$ report truthfully and are rejected; types above u report truthfully and are accepted.
- **Pooling mass comparison** depends on a *bar–boundary race*: derivative $P'(u) = f(u) - f(\theta^*(u))\,\theta^{*\prime}(u)$.
- **Three regimes (talk-up):**
  1. Under stylised priors, pooling *increases* as we move from lenient SM to MoM to strict SM.
  2. Under log-normal (or thin-tailed) densities, pooling increases from lenient SM to MoM and then *decreases* from MoM to strict SM.
  3. Under sufficiently sluggish beliefs (DWBR), pooling *decreases* as we move from lenient SM to MoM to strict SM.
- First two patterns are highly non-obvious given the intuition that "stricter rules should reduce pooling" and the literature treating MoM as uniquely efficient.

Headline Results II: Talk-Down Regime

- **Talk-down equilibrium.**
  - When private benefits of control (PBC) are large enough and single-crossing holds, there is a parallel equilibrium in which:
    - partial pooling occurs *above* the bar: controllers shade valuations down to gain approval; and
    - low types below the marginal threshold report truthfully and are rejected, while high types far above the bar report truthfully and are accepted.
- **Failure of single-crossing.**
  - If single-crossing fails, the marginal type is no longer monotone in project quality.
  - This can generate *multiple equilibria* with non-ordered approval sets, complicating the rule comparison.

Players and Ownership

- One **controlling shareholder** (CS) with voting stake $\alpha \in (0, 1)$.
- Residual stake $1 - \alpha$ held by a large but finite number N of symmetric minority shareholders.
- Each minority holds fraction

$$\lambda := \frac{1}{N}, \qquad \text{so her pro-rata claim on minorities' cash flow is } \lambda(1 - \alpha).$$

- We often speak in terms of a single *representative* minority with exposure λ.

Project Quality and Prior

Assumption 1 (Type distribution and monotone shoulders (P1–P2))

- *Project quality* $\theta \sim F$ *on a compact interval* $[\underline{\theta}_0, \overline{\theta}_0] \subset \mathbb{R}$ *with* $\underline{\theta}_0 < 0 < \overline{\theta}_0$.
- *Density* $f := F' > 0$ *on the support and* $\log f$ *is concave (log–concave prior).*
- *There is a unique mode* u_{MoM} *(later the MoM bar), and*

$$f \text{ nondecreasing on } [\underline{\theta}_0, u_{\mathrm{MoM}}], \quad f \text{ nonincreasing on } [u_{\mathrm{MoM}}, \overline{\theta}_0].$$

Private Benefits and Payoffs

Assumption 2 (Private benefits (P3))

- *Implementing the project yields private benefits* $B(\theta) = b_0 + D(\theta)$ *to the CS.*
- *In the baseline talk–up regime:*

$$0 < B'(\theta) < 1 - \alpha \quad \text{for all } \theta.$$

- *We often specialise to the affine case* $B(\theta) = b_0 + b_1\theta$ *with* $b_1 \in (0, 1 - \alpha)$.
- *Minority payoff from implementation is* $(1 - \alpha)\theta - D(\theta)$, *strictly increasing in* θ.

Signalling Technology

Assumption 3 (Gap-based signalling cost (S1–S4))

- *After observing θ, the CS sends message $m \in \mathbb{R}$.*
- *Cost depends only on the gap $\Delta = m - \theta$:*

$$C(m, \theta) = \eta\,|\Delta| + \frac{\kappa}{2}\Delta^2, \quad \eta > 0,\ \kappa \geq 0,$$

  with $C(\theta, \theta) = 0$.
- *Up–wedge ($m > \theta$): $C_m = \eta + \kappa\Delta > 0$.*
  Down–wedge ($m < \theta$): $C_m = -\eta + \kappa\Delta < 0$.
  So costs are convex and strictly increasing in $|\Delta|$.
- *Symmetry: $C(\theta + \Delta, \theta) = C(\theta - \Delta, \theta)$.*

Beliefs and Updating

Assumption 4 (Posterior mean (M1))

- *For each message m, minorities hold posterior*

$$\mu(m) := \mathbb{E}[\theta \mid m],$$

  which is continuous, strictly increasing, and invertible.
- *Tail regularity:*

$$\lim_{m \to -\infty} \mu(m) = \underline{\theta}_0, \quad \lim_{m \to +\infty} \mu(m) = \overline{\theta}_0.$$

- *On-path beliefs come from Bayes' rule; off-path we use the Grossman–Perry credible-beliefs refinement.*

Minority Utility and Voting

- Each minority i has privately known bias $b_i \sim U[-1, 1]$ (following Levit et al.).
- Conditional on message m,

$$\mathbb{E}[u_i \mid m] = \lambda\big[(1-\alpha)\mu(m) - \beta^{\mathrm{Div}}(m)\big] + b_i,$$

  where $\beta^{\mathrm{Div}}(m) = \mathbb{E}[D(\theta) \mid m]$.
- Investor i votes "Yes" iff $b_i \geq t_\lambda(m)$, where

$$t_\lambda(m) := -\lambda\big[(1-\alpha)\mu(m) - \beta^{\mathrm{Div}}(m)\big].$$

- Fraction of "Yes" votes:

$$\frac{1 - t_\lambda(m)}{2} = \frac{1}{2} + \frac{\lambda}{2}\big[(1-\alpha)\mu(m) - \beta^{\mathrm{Div}}(m)\big].$$

Voting Rules and Belief Cut-off u

- **MoM:** proposal passes iff more than half of minority shares vote "Yes":

$$(1-\alpha)\mu(m) - \beta^{\mathrm{Div}}(m) \;\geq\; 0.$$

- **Supermajority (SM):** overall quota $S > 1/2$. Given controller stake α, minorities must supply

$$\pi := \frac{S-\alpha}{1-\alpha}$$

of minority votes. This yields the unified test

$$(1-\alpha)\mu(m) - \beta^{\mathrm{Div}}(m) \;\geq\; \underline{\kappa}^{(\lambda)}_{\mathrm{rule}},$$

with $\underline{\kappa}^{(\lambda)}_{\mathrm{MoM}} = 0$ and $\underline{\kappa}^{(\lambda)}_{\mathrm{SM}} = -(1-2\pi)/\lambda$.

- For an interior rule we define the **belief cut-off** u and the *minimal persuasive message*

$$m_{\mathrm{req}}(u) := \mu^{-1}(u)$$

by making this inequality bind.

Orientation: Talk-Up vs Talk-Down

- Define $\Phi(m) := (1-\alpha)\mu(m) - \beta^{\mathrm{Div}}(m)$.
- **Talk-up regime (baseline):** $0 < B'(\theta) < 1-\alpha \Rightarrow \Phi$ strictly increasing in μ.
  - Approval iff $\mu(m) \geq u$.
  - Any pooling lies *below* the bar u.
- **Talk-down regime (diversion-dominant):** $B'(\theta) > 1-\alpha \Rightarrow \Phi$ strictly decreasing.
  - Approval iff $\mu(m) \leq u$.
  - Any pooling lies *above* the bar u.

Controller Utility and Lying-Profit

- For a fixed rule (hence fixed u), minority implement iff the passage test holds.
- CS chooses m to maximise

$$U_{\mathrm{CS}}(m,\theta) = \mathbf{1}\{\text{approve}(m)\} \cdot [\alpha\theta + B(\theta)] - C(m,\theta).$$

- **Effective decision:** for each θ, compare truthful rejection vs the *cheapest approved message*, which is $m_{\mathrm{req}}(u)$.
- **Lying-profit (talk-up)**:

$$\Lambda(\theta; u) := \alpha\theta + B(\theta) - C\big(m_{\mathrm{req}}(u), \theta\big).$$

- This function determines whether type θ pools at the bar u or remains truthful.

Result 1: Equilibrium with Talk-Up (Fixed Bar)

Proposition 1 (Talk-up PSE at a fixed bar u)

Work in the talk-up regime $0 < B'(\theta) < 1 - \alpha$ *with Assumptions 1–3 and the gap-based cost. Fix an interior belief cut-off* $u \in (\underline{\theta}_0, \overline{\theta}_0)$ *and define* $m_{\mathrm{req}}(u) = \mu^{-1}(u)$ *and* $\Lambda(\theta; u) = \alpha\theta + B(\theta) - C(m_{\mathrm{req}}(u), \theta)$. *Then:*

- *The function* $\theta \mapsto \Lambda(\theta; u)$ *is strictly increasing on* $[\underline{\theta}_0, u]$.
- *There is a unique boundary* $\theta^*(u) \in [\underline{\theta}_0, u]$ *with* $\Lambda(\theta^*(u); u) = 0$.
- *There exists a Perfect–Sequential Equilibrium (PSE) with*

$$\sigma(\theta) = \begin{cases} \theta, & \theta < \theta^*(u) \quad \textit{(truth; rejected)}, \\ m_{\mathrm{req}}(u), & \theta \in [\theta^*(u), u) \quad \textit{(pool; approved)}, \\ \theta, & \theta \geq u \quad \textit{(truth; approved)}. \end{cases}$$

- *For this* u, *every PSE yields the same implement/reject allocation.*

Talk-Up Equilibrium: Intuition

- On the up-wedge $m_{\mathrm{req}}(u) \geq \theta$, cost derivatives satisfy

$$C_\theta(m_{\mathrm{req}}, \theta) = -(\eta + \kappa\Delta) < 0, \quad \Delta := m_{\mathrm{req}} - \theta \geq 0.$$

- So

$$\partial_\theta \Lambda = \alpha + B'(\theta) - C_\theta \geq \alpha + B'(\theta) + \eta > 0;$$

  lying-profit is strictly increasing in θ.
- Very low types: lying-profit negative, so they prefer truthful rejection.
- Intermediate types: prize from approval outweighs cost, so they shade up to the minimal approved message m_{req}.
- High types $\theta \geq u$: passing is easy at truth, so they separate and pay zero cost.

Pooling Mass and the Bar–Boundary Race

- For a fixed bar u, the pooling set is $[\theta^*(u), u)$.
- **Pooling mass (talk-up):**

$$P(u) := \int_{\theta^*(u)}^{u} f(\theta)\, d\theta.$$

- Differentiate using Leibniz:

$$P'(u) = f(u) - f(\theta^*(u))\, \theta^{*\prime}(u).$$

- The sign of $P'(u)$ depends on the *bar–boundary race*: compare bar speed (weighted by $f(u)$) to boundary speed (weighted by $f(\theta^*)$).

Boundary Responsiveness $\theta^{*\prime}(u)$

From the implicit equation $\Lambda(\theta^*(u); u) = 0$,

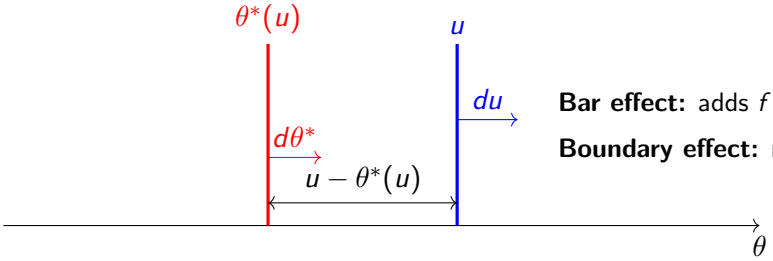
$$\theta^{*\prime}(u) = -\frac{\Lambda_u}{\Lambda_\theta}.$$

Using gap-cost symmetry $C_\theta = -C_m$ on the up-wedge and the definition $m_{\mathrm{req}}(u) = \mu^{-1}(u)$, we obtain

$$\theta^{*\prime}(u) = \frac{\underbrace{C_m\big(m_{\mathrm{req}}(u), \theta^*(u)\big)\,(\mu^{-1})'(u)}_{\text{marginal cost pressure from tightening the bar}}}{\underbrace{\alpha + B'\big(\theta^*(u)\big) + C_m\big(m_{\mathrm{req}}(u), \theta^*(u)\big)}_{\text{total marginal benefit from shifting the boundary}}}.$$

- Numerator: how much the *required lie* becomes more expensive when the bar u increases.
- Denominator: how much the marginal type gains from moving up—via a steeper truthful payoff $\alpha + B'(\theta^*)$ and a smaller lie (cost relief through C_m).

Bar–Boundary Geometry (Talk-Up)



Bar effect: adds $f(u)\,du$ mass

Boundary effect: removes $f(\theta^*)\,d\theta^*$

Sign of $P'(u) = f(u) - f(\theta^*)\,\theta^{*\prime}(u)$ is determined by whether $\theta^{*\prime}(u)$ exceeds the density ratio $r(u) := f(u)/f(\theta^*(u))$.

Result 2: Uniform Prior & Identity Beliefs

Assumption 5 (Uniform prior & identity beliefs)

- *Prior uniform on* $[a, b]$*; density* $f(\theta) = 1/(b - a)$.
- *Identity beliefs:* $\mu(m) = m$.
- *For each interior bar* u*, the minimal persuasive message is* $m_{\mathrm{req}}(u) = u$.

Result 2: Uniform Prior & Identity Beliefs

Proposition 2 (Uniform & identity: pooling rises with the bar)

On any corridor where partial pooling persists, we have $P'(u) > 0$, so pooling mass increases with the bar. If the rule cut-offs satisfy

$$u_{\mathrm{SM}}^{\mathrm{len}} < u_{\mathrm{MoM}} < u_{\mathrm{SM}}^{\mathrm{str}},$$

then

$$P\big(u_{\mathrm{SM}}^{\mathrm{len}}\big) \;<\; P\big(u_{\mathrm{MoM}}\big) \;<\; P\big(u_{\mathrm{SM}}^{\mathrm{str}}\big).$$

Uniform Prior: Intuition

- Uniform density: $f(u) = f(\theta^*(u))$.
- With identity beliefs $\mu(m) = m$ and gap costs, the boundary slope is

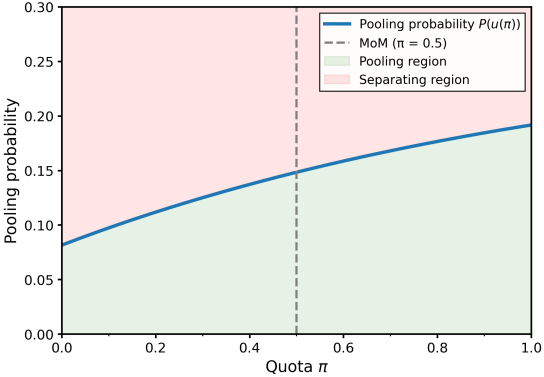
$$\theta^{*\prime}(u) = \frac{C_m\big(u, \theta^*(u)\big)}{\alpha + B'\big(\theta^*(u)\big) + C_m\big(u, \theta^*(u)\big)},$$

  so $0 < \theta^{*\prime}(u) < 1$.
- Bar–boundary derivative:

$$P'(u) = f(u) - f(\theta^*)\,\theta^{*\prime}(u) = f(u)\big(1 - \theta^{*\prime}(u)\big) > 0.$$

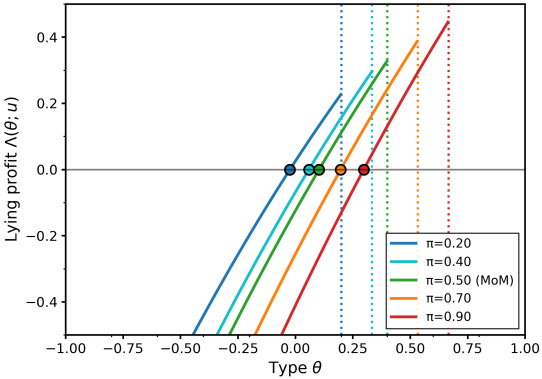
- Economic reading:
  - Tightening the bar increases the lie required to pass.
  - Some marginal types drop out, but the induced movement in the boundary is *less than one-for-one*, so the pooling interval widens.

Uniform Prior: Simulated Pooling Probabilities



Illustrative pooling probability $P(u(\pi))$ under a uniform prior and identity beliefs. As the quota π (and thus the bar u) rises, pooling mass increases: lenient SM $\rightarrow$ MoM $\rightarrow$ strict SM.

Uniform Prior: Lying-Profit Curves



Each curve shows $\Lambda(\theta; u)$ for a different quota. The zero-crossing pins down $\theta^*(u)$; as the bar increases, the boundary moves right more slowly than the bar, so the pooling band $[\theta^*(u), u]$ expands.

Result 3: Log-Concave Prior with Thin Tails

Assumption 6 (Centered log-concave prior with thin tails)

- *Identity beliefs* $\mu(m) = m$.
- *Prior* f *log–concave with unique mode at the MoM bar* $u_{\mathrm{MoM}} = 0$.
- *Left shoulder:* f *(weakly) increasing; right shoulder:* f *decreasing.*
- **Thin right tails:** *on the right corridor, the density ratio* $r(u) := f(u)/f(\theta^*(u))$ *is uniformly small, so the bar always sits in a thinner region than the boundary.*

Result 3: MoM Maximises Pooling

Proposition 3 (MoM maximises pooling under thin tails)

Under the assumptions above and along corridors where partial pooling persists, suppose the right shoulder is thin in the sense that

$$\sup_{u \in [0,\, u_{\mathrm{SM}}^{\mathrm{str}}]} \frac{f(u)}{f(\theta^*(u))} \le \underline{\gamma} \quad \text{with} \quad \underline{\gamma} := \frac{\eta}{\alpha + \overline{B'} + \eta} \in (0,1), \quad \overline{B'} := \sup_{\theta} B'(\theta).$$

Then:

- *For $u < 0$ (left shoulder), pooling mass $P(u)$ strictly increases as $u \uparrow 0$.*
- *For $u > 0$ (right shoulder), we have $P'(u) \le 0$: pooling decreases as u moves right.*

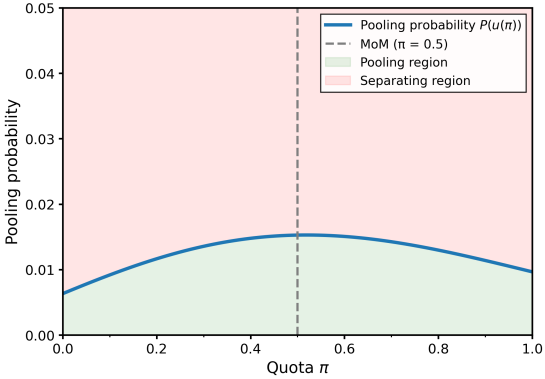
Consequently, among all interior bars in the pooling corridors,

$$P(u) \text{ is maximised at } u = 0 = u_{\mathrm{MoM}}.$$

Centered Log-Concave: Intuition

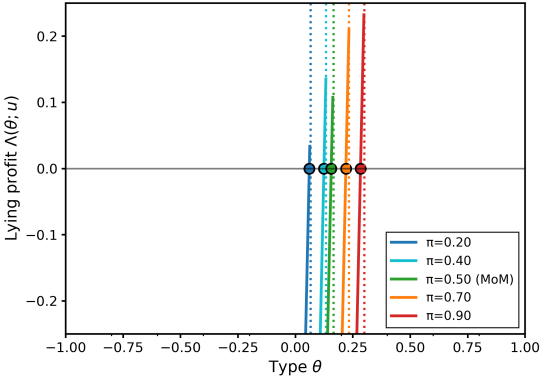
- On the **left shoulder**: $f(u) \geq f(\theta^*(u))$ and $0 < \theta^{*\prime}(u) < 1$, so $P'(u) > 0$: tightening towards MoM increases pooling.
- On the **right shoulder**: $f(u) \leq f(\theta^*(u))$ and thin tails imply $\theta^{*\prime}(u)$ dominates the density ratio $\Rightarrow P'(u) \leq 0$: tightening beyond MoM reduces pooling.
- MoM sits at the mode, where the pooling band lies in the "thickest" region of the prior; tighter or looser rules shrink the pooling band.

Centered Log-Concave: Simulated Pooling Probabilities



With a centred log-concave prior (e.g. truncated normal) and identity beliefs, pooling probability peaks at the MoM quota: lenient and strict supermajorities both induce less pooling than MoM.

Centered Log-Concave: Lying-Profit Curves



As the quota moves away from MoM, the lying-profit curves tilt so that the zero-crossing moves closer to the bar, shrinking the pooling interval.

Density-Weighted Boundary Responsiveness (DWBR)

- Recall $P'(u) = f(u) - f(\theta^*)\,\theta^{*\prime}(u)$.
- Define the density ratio $r(u) := \dfrac{f(u)}{f(\theta^*(u))}$.
- **DWBR condition (talk-up):**

$$\theta^{*\prime}(u) \;\geq\; r(u) \quad \text{for all interior } u \text{ on a corridor.}$$

- With *sluggish beliefs* $\mu(m) = Km$ with $K \in (0, 1]$ and affine benefits $B(\theta) = b_0 + b_1\theta$, gap costs give

$$\theta^{*\prime}(u) = \frac{\eta + \kappa\Delta(u)}{\alpha + b_1 + \eta + \kappa\Delta(u)} \cdot \frac{1}{K}, \qquad \Delta(u) := m_{\mathrm{req}}(u) - \theta^*(u) > 0,$$

so

$$\theta^{*\prime}(u) \;\geq\; \frac{\eta}{K(\alpha + b_1 + \eta)}.$$

- **Sufficient condition for global DWBR:** if

$$\frac{\eta}{K(\alpha + b_1 + \eta)} \;\geq\; \sup_{u \text{ on corridor}} \frac{f(u)}{f(\theta^*(u))},$$

then DWBR holds on that corridor.

Result 4: Global DWBR Ordering (Talk-Up)

Proposition 4 (Global DWBR: pooling strictly decreases with the bar)

Suppose that on both pooling corridors the DWBR condition holds:

$$\theta^{*\prime}(u) \;\geq\; \frac{f(u)}{f(\theta^*(u))} \quad \textit{for all interior } u$$

between the lenient SM bar and the strict SM bar. Then

$$P'(u) \leq 0$$

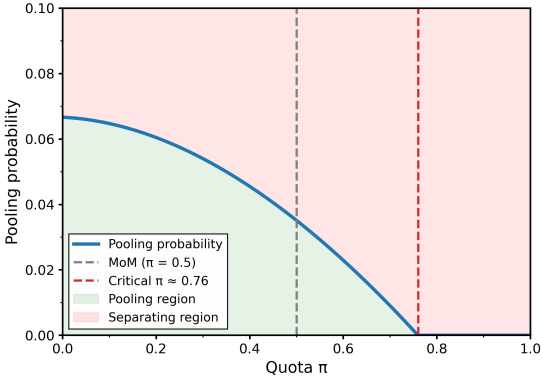
on each corridor. If the rule cut-offs satisfy $u_{\mathrm{SM}}^{\textit{lenient}} < u_{\mathrm{MoM}} < u_{\mathrm{SM}}^{\textit{strict}}$, *we obtain the reversed ranking*

$$P\big(u_{\mathrm{SM}}^{\textit{lenient}}\big) \;>\; P\big(u_{\mathrm{MoM}}\big) \;>\; P\big(u_{\mathrm{SM}}^{\textit{strict}}\big).$$

DWBR Ordering: Intuition

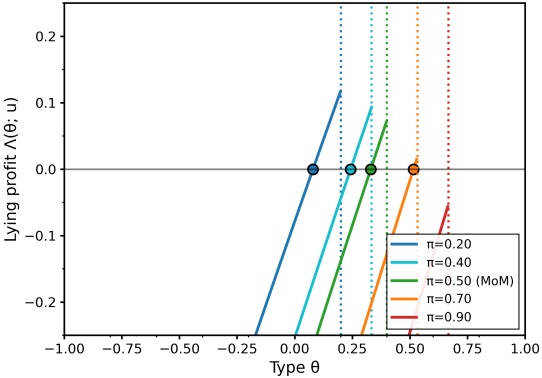
- Under DWBR, the *behavioral* response of the marginal type is so strong that the boundary chases the bar in density-weighted units.
- When beliefs are sluggish ($K < 1$), minorities underreact to messages, so CSs must lie more to pass. This makes $\theta^{*\prime}(u)$ large.
- Tightening the bar increases the lie burden enough that many marginal types drop out of pooling; the pooling band shrinks.
- Result: lenient SM has the greatest pooling; MoM intermediate; strict SM the least pooling.

DWBR: Simulated Pooling Probabilities



Under sluggish beliefs and parameter values satisfying DWBR, pooling probability decreases in the quota. Past a critical strict quota, pooling vanishes and the equilibrium separates.

DWBR: Lying-Profit Curves



As the quota tightens under DWBR, the lying-profit curves shift so that the zero-crossing $\theta^*(u)$ moves outward rapidly relative to the bar; the pooling interval collapses.

Talk-Down Regime

- When $B'(\theta) > 1 - \alpha$, private benefits grow so steeply that minorities prefer lower perceived value.
- Approval test reverses: *approve iff* $\mu(m) \leq u$.
- For $\theta > u$, the *least-cost approved message* is now the *largest* message with $\mu(m) \leq u$, i.e. $m_{\mathrm{req}}(u) = \mu^{-1}(u)$.
- Define the talk-down lying-profit

$$\Lambda^{\downarrow}(\theta; u) := \alpha\theta + B(\theta) - C(m_{\mathrm{req}}(u), \theta).$$

Result 5: Equilibrium with Talk-Down (Single Crossing)

Talk-down single-crossing assumption. Work in the diversion-dominant regime $B'(\theta) > 1 - \alpha$. For $\theta \geq u$, define the talk-down lying-profit

$$\Lambda^{\downarrow}(\theta; u) := \alpha\theta + B(\theta) - C\big(m_{\mathrm{req}}(u), \theta\big), \qquad m_{\mathrm{req}}(u) := \mu^{-1}(u).$$

A sufficient condition for strict single-crossing on $[u, \overline{\theta}_0]$ is

$$\alpha + \sup_{\theta \in [u, \overline{\theta}_0]} B'(\theta) \;\leq\; \eta,$$

which implies $\Lambda^{\downarrow}_{\theta}(\theta; u) < 0$ there and hence a unique zero $\theta^{\downarrow}_{*}(u) \geq u$.

Proposition (talk-down PSE). Fix an interior bar u and assume the above single-crossing. Then there exists a PSE with

$$\sigma(\theta) = \begin{cases} \theta, & \theta \leq u \quad \text{(truth; approved)}, \\ m_{\mathrm{req}}(u), & \theta \in (u, \theta^{\downarrow}_{*}(u)) \quad \text{(pool above } u\text{; approved)}, \\ \theta, & \theta \geq \theta^{\downarrow}_{*}(u) \quad \text{(truth; rejected)}. \end{cases}$$

In any PSE with this bar u, the project is implemented iff $\theta \leq \theta^{\downarrow}_{*}(u)$.

Talk-Down Equilibrium: Intuition

- Now minorities dislike high values of θ, because diversion grows faster than their pro-rata share.
- Types just above u can *talk down* cheaply to $m_{\mathrm{req}}(u)$ and get approved.
- For sufficiently high θ, the diversion is so large that even the minimal talk-down lie is not worth paying for; truthful rejection dominates.
- Single-crossing ensures a unique upper boundary $\theta_*^{\downarrow}(u)$ between these regions.

Talk-Down Ordering Under DWBR

Define the upper pooling mass

$$Q(u) := \int_u^{\theta_*^{\downarrow}(u)} f(\theta)\, d\theta,$$

so pooling lies *above* the bar.

Talk-down DWBR assumption. On corridors of partial pooling, the upper boundary satisfies

$$\theta_*^{\downarrow\prime}(u) \;\geq\; \frac{f(u)}{f\big(\theta_*^{\downarrow}(u)\big)} \quad \text{for all interior } u \text{ on the corridor.}$$

Talk-Down Ordering Under DWBR

Proposition 5 (Talk-down DWBR ordering)

Suppose the rule cut-offs satisfy $u_{\mathrm{SM}}^{\textit{strict}} < u_{\mathrm{MoM}} < u_{\mathrm{SM}}^{\textit{lenient}}$ *in the talk-down regime and talk-down DWBR holds on both corridors. Then*

$$Q\big(u_{\mathrm{SM}}^{\textit{lenient}}\big) \;\geq\; Q\big(u_{\mathrm{MoM}}\big) \;\geq\; Q\big(u_{\mathrm{SM}}^{\textit{strict}}\big),$$

with strict inequalities generically.

Multiple Equilibria: Talk-Down without Single-Crossing

Proposition 6 (Talk-down without single-crossing)

In the diversion-dominant (talk-down) regime, fix a rule that induces an interior bar u and let $m_{\mathrm{req}}(u) = \mu^{-1}(u)$. Define the talk-down lying-profit

$$\Lambda^{\downarrow}(\theta; u) := \alpha\theta + B(\theta) - C\big(m_{\mathrm{req}}(u), \theta\big).$$

If $\theta \mapsto \Lambda^{\downarrow}(\theta; u)$ is not single-crossing on $(u, \overline{\theta}_0]$, then the set

$$\mathcal{A}(u) := \{\theta > u : \Lambda^{\downarrow}(\theta; u) \geq 0\}$$

need not be an interval: it can be a finite union of disjoint intervals. There exist:

- *a* **maximal-pooling** *PSE in which all $\theta \in \mathcal{A}(u)$ pool at $m_{\mathrm{req}}(u)$; and*
- *a* **minimal-pooling** *PSE in which no type above u pools (truthful rejection for all $\theta > u$),*

supported by different GP-credible off-path beliefs.

Result 6: Multiple Equilibria: Intuition

- Without upper-side single-crossing, $\Lambda^{\downarrow}(\theta; u)$ can cross zero multiple times, so profitability of talk-down can alternate as θ increases.
- The least-cost approved message is always $m_{\mathrm{req}}(u)$, so any type in $\mathcal{A}(u)$ would like to join any on-path pool at $m_{\mathrm{req}}(u)$.
- With optimistic beliefs, all profitable types are drawn into the same pool (maximal pooling).
- With sufficiently pessimistic GP-credible beliefs at off-path messages, it can be an equilibrium that no upper-side type talks down (minimal pooling).
- Comparative statics in u are no longer monotone: the rule-induced bar can flip which pockets of types are approved.

Conclusion + Extensions

- Voting thresholds affect not just pass/fail probabilities but the *equilibrium communication technology*.
- Baseline talk-up with responsive beliefs:
  - Uniform prior: pooling smallest under lenient SM, largest under strict SM.
  - Thin-tailed log-concave prior: MoM can *maximise* pooling.
- With sluggish beliefs and DWBR, ordering flips: pooling largest under lenient SM, smallest under strict SM.
- In talk-down environments, analogous DWBR logic produces an ordering for upper pooling mass.

Conclusion + Extensions

- **Endogenous ownership α.** Controllers choose their stake anticipating how MoM vs. lenient/strict SM affect disclosure incentives and approval probabilities.
- **Strategic minorities / information acquisition.** Allow minority investors to acquire costly information or coordinate, rather than behaving as purely passive Bayesian updaters.
- **Empirical tests.** Exploit cross-jurisdictional or quasi-experimental changes in MoM and supermajority rules to test predictions on pooling, disclosure quality, and approval rates.